# Ashutoshh Singh

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Databricks Certified Data Engineer Associate

## Professional Summary

I am a Senior DevOps and Data Engineer with over 15 years of industry experience, specialising in cloud-native infrastructure, container orchestration, data engineering, and AI-driven platforms.As a corporate trainer and consultant, I have worked directly with CXOs, Engineering Directors, and senior IT teams at organisations including Cisco, Oracle, E&Y, JPMC, and Walmart,delivering measurable improvements in engineering capability and system performance.

## Industry Experience as Corporate Trainer: 15+ Years

### Co-Founder & CEO **Delvex Innovations Pvt Ltd**

- AI & Deep Learning Specialist with strong expertise in NLP and applied AI solutions
- Certified Databricks Instructor associated with NIIT — enterprise data engineering & analytics training
- Red Hat Certified Trainer — enterprise Linux, container platforms, and OpenShift
- DevOps Engineer — CI/CD pipelines, containerisation, infrastructure automation, cloud-native deployments
- Cisco Networking Expert — IOS, NX-OS, IOS-XE, IOS-XR, NFV & SDN

**Co-Founder | CTO at Learn and Build**

Working here for Course Content Creation & Trainer Engagement & Task Deployment for Projects Execution and Assignment Related Works.

**Founder | CTO at Adhoc Networks Infotech Private Limited**

Working here as Sr. Technical Trainer, Hadoop Analyst, DevOps Engineer and AWS Expert, OpenStack Cloud Developer. Working in the field of Data Science on implementation of Deep Learning & Machine Learning in Automation Ansible-Containers, Kubernetes with app hosting, Cloud foundation and Cloud foundry with DevOps, Jenkins, Puppet.

**DevOps Engineer, R&D and Sr. Trainer at Confidential**

Served as OpenStack Cloud Administrator and Linux Operations Engineer, maintaining the cloud infrastructure within the company. Delivered training and maintained fully updated and functional testing environment on technologies like oVirt, OpenStack, Apache Hadoop, GluterFS, Cisco Switches, Routers and Firewalls.

**Linux System Administrator & AWS Cloud Engineer at WIPL**

Managed web hosting for the company using WHM and Cpanel. Maintained multiple mail servers, using Sendmail, Postfix and Exim.

## List of prominent Trainings Delivered to Clients:

| | Clients Technologies |
|---|---|
| **Walmart, Bengaluru ● Linux** | ● **Graphite**<br>● **Grafana**<br>● **AI** |
| **Wipro, Mysore ● Redis Admin** | ● **AWS Redshift**<br>● **App Dynamics**<br>● **Kong** |
| **Software AG, Bengaluru ● Docker**<br><br>**PWC ● Redis** | ● **Certified Kubernetes Application Developer CKAD** |

**Mercedes Benz - India ● Linux**

- **OpenStack**

**Oracle, Bengaluru ● Docker**

- **Advanced Kubernetes**
- **AppDynamics**
- **AI**
- **Git**
- **Jenkins – CI/CD**
- **Splunk Admin**
- **DevSecOps**
- **Kong**

- **Redis**

**ATOS, Bengaluru ● Hubstor – Cloud Data Management tool**

**Fidelity, Bengaluru ● Docker and Linux Containers ● Shell Scripting**

- **Python**
- **R**
- **Perl**

**Cisco, Bengaluru ● Python**

- **Machine Learning**
- **Network Automation**
- **Docker with Cisco IOS**
- **AppDynamics**
- **Docker**
- **Kubernetes**
- **DataDog on DevOps**
- **Splunk Admin**
- **Redis**
- **AWS CloudFormation**

**JP Morgan, Bengaluru ● Delphix**

- **Unix**

- **AI**

**Infosys, Hyderabad ● Rancher**

**CareerSocially, Noida ● Machine Learning & NLP ● Image & Speech Recognition**

- **TensorFlow, Open CV**
- **AI**

**Cognizant, Noida ● Deep Learning & Image Processing ● OpenShift Application Development**
**● AppDynamics**

**iNurture, Bengaluru ● TensorFlow & OpenCV ● App Dynamics**
**● Neural Networks**
**● Big Data Analytics**

**Vani, Gurugram ● Deep Learning - Phrase Extraction & Speech recognition**
**● AWS Redshift**

**Altran - Aricent, Bengaluru ● Puppet**
**● Ansible**
**● Chef**
**● Salt Stack**
**● Redshift**

**Mphasis, Bengaluru**
**● CA test Data management**
**● Ansible**

Mode:

**● DevOps Infra Provisioning**

Online Mix Batch

**● CloudFormation**

Classroom Batch

**● Nginx**
**● Chef**
**● Neural Networks**

**Honeywell, Bengaluru ● OpenShift Origin**

**Ericsson, Gurugram ● Rubrik – Dataos IO product for No SQL database**
**● Amazon Redshift**
**● Neural Networks & AI**

**Fidelity, Bengaluru CoreOS and rocket with Google Kubernetes Splunk Administration**

**Fidelity, Chennai ● AWS Sol Architect**
**● Elastic Search**
**● RabbitMQ**
**● AI**

**Johnson Controls India, Pune ● Jenkins and DevOps Tools ● Maria DB**
**● Nginx**
**● Big Data Analytics**
**● Distributed System Architecture**

**E&Y, Bengaluru ● DevOps**
**● Nginx**
**● App Dynamics**
**● Kubernetes**
**● Terraform**
**● NLP, TensorFlow, Open CV**
**● Splunk Administration**

**Capgemini, Bengaluru ● DevOps Fundamentals ● Prometheus**
**● Nginx**
**● Unix**
**● GPU Programming Model**

**CGI, Bengaluru ● Storage solution and Management ● Ansible**
**● Maria DB**
**● OpenShift**
**● Splunk Administration**

**BFC Datasystems, Bengaluru ● Cloud Foundry**
**● OpenStack**
**● Google Cloud Platform**
**● Prometheus**
**● OpenShift**

**Epsilon, Bengaluru ● AWS Fundamentals**
**● Grafana**
**● OpenStack Administrator**

**West Corporation, Bengaluru ● AWS - Lambda**

- **SQS**
- **SES**
- **DynamoDB**
- **API Gateway and ElasticSearch with Spark**

**TCS, Bengaluru ● RedHat OpenStack Administration III**

- **Red Hat Ansible Automation**
- **Maria DB**

**Adobe, Noida ● Splunk Administration**

**Adhoc Networks – Red Hat Partners DO280 -** Red Hat OpenShift Administration I **DO288 -** Red Hat OpenShift Development I: Containerizing Applications
**DO407 -** Automation with Ansible
**DO409 -** Automation with Ansible II: Ansible Tower **RH415 -** Red Hat Security: Linux in Physical, Virtual, and Cloud
**RH413 -** Specialist in Server Security and Hardening **RH362 -** Red Hat Security: Identity Management and Active Directory Integration
**RH318** - Red Hat Virtualization Administrator
**RH236 -** Red Hat Gluster Storage Administrator
**RH342 -** Red Hat Enterprise Linux Diagnostics and Troubleshooting

**CEPH125 -** Red Hat Ceph Storage Architecture and Administration
**CL110 -** Red Hat OpenStack Administration I: Core Operations for Cloud Operators
**CL210 -** Red Hat OpenStack Administration II: Infrastructure Configuration for Cloud Administrators
**RH294 -** Red Hat System Administration III: Linux Automation with Ansible
**RH254 -** Red Hat System Administration III: Data Center Services for Red Hat Enterprise Linux 7

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|---|---|
| **Adhoc Networks Infotech Pvt Ltd ● Data Science Specialization in Machine Learning** | **● AWS ML Model, Chatbots, NLP Advance**<br>**● Matplotlib, R programming**<br>**● Data Analytics Specialization in Big Data**<br>**● Apache Hadoop & Apache Spark, Spark SQL**<br>**● Databricks, Azure Datalakes, Yarn, Mapreduce, Power BI, Tableau**<br>**● Ansible, Terraform, Jenkins, Version Control, LAMP**<br>**● Deep Learning: OpenCV, Tensorflow, Keras, Numpy, Image processing, Face Detection**<br>**● Advanced AI with NLP & Neural Networks Python Programming, PyTorch, CGI, Django, Flask**<br>**● Cloud – AWS & Azure**<br>**● Grafana** |
| **Decurtis, Jaipur ● Cloud Computing with AWS ● Digital Ocean and Azure** | |
| **Celebal Technologies, Jaipur ● Big Data Hadoop & Spark ● Databricks & Azure Datalakes** | |
| **Sapient, Noida ● Red Hat OpenStack Administrator CL210** | **● Kubernetes - CKA Module**<br>**● Jenkins, GIT, Microservices** |
| **GIPL, Jaipur ● Cisco Certified Network Administrator (Routing and Switching, Security** | **● RedHat Certified Specialist in Server Hardening & Security)** |

| | |
|---|---|
| **TCS, Pune**<br><br>WebEx & Classroom | **● Ansible Tower**<br>**● Nagios** |

## Professional Certifications [Red Hat Certification ID: 120-053-087]

- Red Hat Certified Architect in Infrastructure [RHCA]
- Red Hat Certified Specialist in Ansible Automation [EX407]

- Red Hat Certified Specialist in OpenShift Administration [EX280]
- Red Hat Certified Specialist in Server Security and Hardening [EX413]
- Red Hat Certified System Administrator in RedHat OpenStack [EX210]
- Red Hat Certified System Administrator in RedHat Gluster Storage Administration [EX236]
- Red Hat Certified System Administrator in Red Hat Enterprise Linux 8 [EX200]
- Red Hat Certified Engineer in Red Hat Enterprise Linux 8 [EX294]
- Red Hat Certified System Administrator in Red Hat Enterprise Linux 7 [EX200]
- Red Hat Certified Engineer in Red Hat Enterprise Linux 7 [EX300]
- Red Hat Certified System Administrator in Red Hat Enterprise Linux 6 [EX200]
- Red Hat Certified Engineer in Red Hat Enterprise Linux 6 [EX300]
- Cpanel & WHM Administrator Certification
- Linux Foundation Certified Kubernetes Administrator (CKA)
- Linux Foundation Certified Kubernetes Application Developer (CKAD)
- Linux Foundation Certified SysAdmin (LFCS)
- Linux Foundation Certified Kubernetes Security (CKS)
- Linux Foundation Certified Engineer (LFCE)

## Technical Domains Expertise for Corporate Training Deliveries

| DevOps | Data Science & Machine Learning |
|---|---|
| ● Ansible | ● Apache Beam |
| ● Puppet | ● Apache Kafka |
| ● Chef | ● Apache Mahout |
| ● Terraform | ● Apache Flint |
| ● Salt Stack | ● Apache Spark |
| ● Docker | ● Big Data |
| ● Kubernetes | ● Cloudera & Apache Hadoop |
| ● Jenkins | ● Yarn |
| ● Team city | ● RabbitMQ |
| ● Prometheus | ● Hive |
| ● GitHub | |

- Artifactory - SDLC
- Microservices
- Monitoring Tools – Nagios, Grafana, Cacti
- Maven
- DataDog
- AiOPS
- Splunk
- Elastic Search
- Sqoop
- HBase & Cassandra
- NLP
- TensorFlow & Keras
- Machine | Deep Learning & AI
- Sci-kit Learn
- Computer Vision
- Neural Networks
- Data Visualization
- Data Bricks

- Logstash
- Kibana
- Power BI

**Programming**

- Python
- R
- Scala
- Go
- Bash – Shell Scripting
- Perl
- YAML

**Networking**

- Cisco R&S
- Cisco Network Security
- Cisco SDN
- Cisco Firewalls
- Cisco VPN
- Cisco UCS
- Juniper Networking

| RedHat Tracks | Cloud Platforms - OpenStack |
| --- | --- |
| ● RHCE<br>● Server Hardening & Troubleshooting<br>● Ansible & Ansible Tower<br>● CEPH & Gluster Storage<br>● OpenShift<br>● Virtualization<br>● JBoss<br>● OpenStack<br>● Performance Tuning | ● NOVA<br>● Neutron<br>● Cinder<br>● Swift<br>● Glance<br>● Ironic<br>● Keystone<br>● Horizon<br>● Manila<br>● Heat |
| **Amazon Web Services - AWS** | **Microsoft Azure** |
| ● Solution Architect – [Associate \| Professional]<br>● SysOps<br>● Developer<br>● Amazon Security<br>● Lambda Services<br>● Amazon Machine Learning<br>● Amazon Data Science | ● Azure Virtual Machines<br>● Azure active directory services<br>● Azure Cloud<br>● Azure Microservices<br>● Azure Blob storage<br>● Azure Databricks<br>● Azure Networking<br>● Azure Load Balancer<br>● Azure Data Lakes & Databricks |

**PingFederate - Ping Identity**

- Working with IDM system like AD / LDAP & Kerberos
- SAML experience
- Working with Oauth 2.0 protocol OpenID expertise & security token
- Configuration and authentication
- Clustering in PingFederate

**PingAccess - Ping Identity**

- Worked with ping access component
- Identity mapping and virtual hosting Agent installation and sites
- Application rewrite rules
- WAM and API access control rules
- SSL certificate and Trust stores

## Other Technology Expertise

- Tools CPanel
- WHM
- App Dynamics
- Webim
- Nagios
- Wireshark
- tcpdump
- xplico
- Burp Suite
- yersinia
- Metasploit
- Servers Apache
- Nginx
- Squid
- NIS
- NFS
- FTP
- Telnet
- SSH
- DHCP
- Log server

| | |
|---|---|
| ● SELinux<br>● PAM<br>● AIDE<br>● Firewall<br>● TCP Wrappers<br>● ACL<br>● Linux Advanced Permissions<br>● Cgroups<br>● Kernel patches<br>● LUKS | ● Kickstart<br>● SAMBA<br>● DNS<br>● Kerberos<br>● NTP<br>● iSCSI<br>● LDAP<br>● targetcli<br>● Unbound DNS |
| ● Mail Servers Postfix<br>● Sendmail<br>● Exim | ● RAID<br>● VMware<br>● ESXi |
| ● MySQL Database<br>● MariaDB<br>● PostgreSQL<br>● NoSQL<br>● MongoDB<br>● RedShift | ● Network boot<br>● PXE<br>● Kickstart<br>● NfSen<br>● Bluecoat Packet Shaper |
| ● Networking Cisco Devices:<br>● VLANs, VRF, STP, SNMP, VTP, Routing, VPN, Switching | ● Security Cisco ASA firewalls<br>● Linux firewalls<br>● Windows firewalls |

- Virtualization oVirt, KVM
- RHEV
- Oracle Virtualbox
- Citrix Xen

## Consulting Assignments

- Worked on version control and revision control system like GIT,SVN. ● Worked LXC, docker and containers technologies with CoreOS.
- Deployed Container clustering management using Swarm and Google Kubernetes. Experience in CI/CD tools like Jenkins, Hudson.
- Deploy our OWN cloud and Architect each Service by own code using Python like Saas, Staas, Naas, Paas, Iaas
- Setup of all Amazon AWS Products like EC2, VPC, ELB, S3, Glacier, EMR, DynamoDB, EBS,RDS, IAM, many more
- Worked as Hadoop administrator track using Bigdata Hadoop with framework like PIG ,HIVE , HBASE, Apache ZooKeeper, Flume , Sqoop etc.
- Worked with Apache Mahout in pattern analysis using map reduce for various use cases being collaborative filtering, clustering, classification & mining.
- Worked and deployed NqSQL and Cluster-based relational database using Hbase ,mariadb to manage the structured and unstructured database. Written many MLS and MCS policies for managing SELinux, for managing webapps and websites.
- Managing and implementing OpenStack Mulitnode cluster with Nova, Neutron, Cinder, Glance, Swift, Qpid, Keyston and many more.
- Set and Troubleshoot many service on AWS cloud like EC2, RDS, VPC, S3, EBS, AMI, IAM, Elastic Beanstalk, Beanstack, EC2 container, EMR, Dynamodb, CloudFormation, Route 53 & many more.
- Designed web front based on CGI scripting using python and Docker API for Packaging of Containers.
- High level automation using Python, Bash, YAML, Perl.
- Data Analysis with R language over Hadoop.
- Apache spark for Hadoop processing.
- TeamCity, Cloud Foundry, Rancher and OpenShift.
- Kernel Development and Microservices.
- Courseware Development for AdHoc Networks, Jaipur

## Research Projects

**Data leakage prevention using Cloud Computing:**

One click data loss prevention and backup mechanism using RAID, Python on RHEL 6, whereby the consumer does not worry anymore about power outages or hardware failure, as all his data is being replicated and backed up in the cloud.

**Automated bug fixing and kernel parameter management:**

In a situation when a bug is released for the current kernel version but there is no sysadmin available, our tool will either apply the patch automatically or sets the kernel parameters such that the bug is no more exploitable, based on the severity of the bug, alerting the IT team in the meanwhile.

**APM Tool management using App Dynamics:**

- Troubleshooting & Monitoring of Application
- Clustering Technique in App Dynamics
- Proactive Monitoring & Health Check of Logs
- Health check of Microservices & Analysis

**Global Email Address Verification Tool:**

The tool takes input from any list, extracts all the email addresses from it and processes them in a manner that all the email addresses are verified over remote connection to email servers of the respective email address and reported to the user whether they are valid or not.

**Voice Controlled Bots - AI**

- Automation of DevOps Tools like Docker & Kubernetes
- Automation of Cloud such as AWS & Azure
- Voice controlled Face recognition
- Data Analysis using Amazon SageMaker, Splunk ML, Databrick, MLib
- AWS Rekognition

**Big Data Hive over Apache Hadoop using IaaS Cloud:**

- Setup Complete Big Data Hadoop Technology of Multi Node cluster with Hbase cluster
- Manage Each and Every Component of big data for distributed storage HDFS, distributed computing MapReduce, Distributed Scheduling YARN
- Works on Big Data Hadoop Related Framework like PIG, HIVE, HBase, Oozie, Zookeeper, Flume, Sqoop, Splunk many more.

**Container Clustering management with Ansible:**

- Developed an auto launched container cluster using Swarm and Kubernetes with Ansible to deploy real time application
- PAAS implementation with Openshift and Cloud foundry.